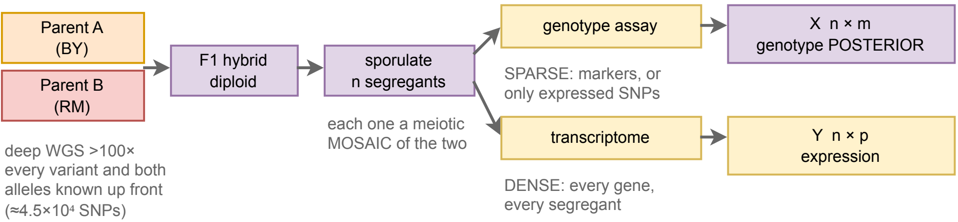
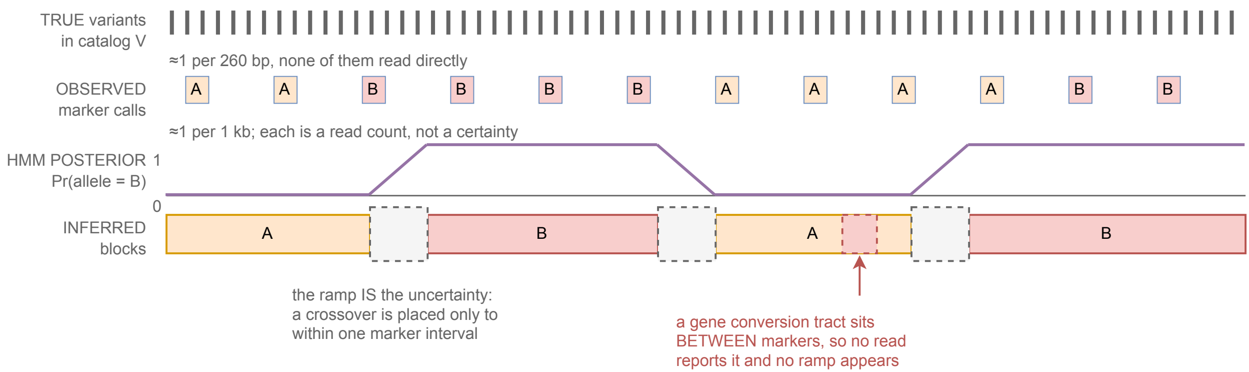
a The cross, and the two things measured on every segregant



The asymmetry. The phenotype is read directly. The genotype never is: it is INFERRED from sparse observations as a mosaic of two known genomes. Panel b is that inference; panels d and e are what it costs.

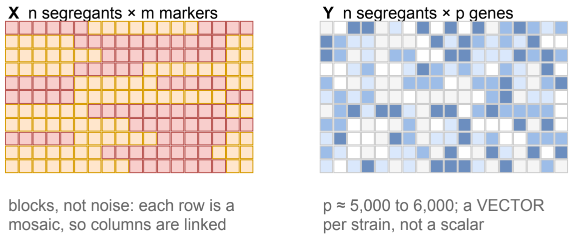
b One segregant, one chromosome: from sparse reads to a posterior



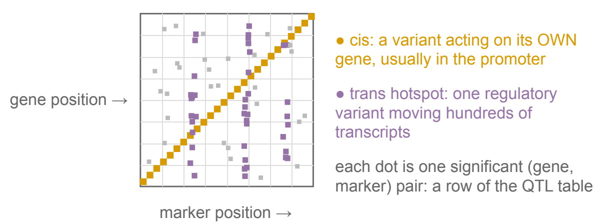
Why sparse reads still suffice. The segregant is never sequenced de novo. At each locus it is ASSIGNED to one of two already-sequenced genomes. Meiosis makes ≈90 crossovers, so a genome is ≈10² blocks, and one marker fixes every cataloged variant in its block. The task is locating ≈10² boundaries, not determining ≈10⁴ values.

What no marker density fixes. Boundary intervals leave ≈0.8% of cataloged sites ambiguous. Non-crossover gene conversion (≈46 per meiosis, ≈2 kb) affects more bases than that and is invisible to sparse markers. De novo mutation, aneuploidy (a haploid can carry a disomy) and non-reference sequence sit outside the mosaic model entirely.

c What the two matrices look like



d The resulting map: cis on the diagonal, trans in bands



e What a record stores, as built for the Bloom 2019 segregant panels

Stored. One record per segregant per condition, (genotype, environment) → phenotype. The genotype is a SegregantGenotype, a sibling of Genotype rather than a perturbation leaf: HaplotypeBlocks of (chromosome, start, end, parent, posterior, n_markers) against two SegregantParents pinned to sha256-anchored assemblies. The variant set is a derived view. Measured: 83 to 143 blocks per segregant (medians by cross), tails to 1,847 from hard calls at posterior 1.0.

Not stored, and what stays open. The QTL table is an estimate conditional on method, threshold and marker density, with no source value to check, so it is not a Phenotype. A mosaic needs no systematic_gene_name, since it is not a GenePerturbation; a cis-eQTL is still usually an intergenic promoter variant with no gene to attach to, so per-variant attribution and the reporter-assay class stay one open decision.